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# A Comparative Analysis of the Prevalence of Artificial Intelligence and Plagiarism in Doctoral Dissertations Using Turnitin

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**Abstract**
Academic misconduct is a prevalent issue in higher education with detrimental effects on the individual students, rigor of the program, and strength of the workplace. Recent advances in artificial intelligence (AI) have reinvigorated concern over academic integrity and the potential use and misuse of AI. However, there is a lack of research on academic integrity in doctoral dissertations. Consequently, the purpose of this study is to explore academic misconduct in dissertations, specifically investigating the prevalence of AI use and plagiarism. Considering debate over the accuracy of technology in detecting AI-generated text, dissertations were also analyzed from before widespread AI availability, as a point of comparison. A sample of 200 dissertations from 2013 and 2023 were analyzed through Turnitin to flag plagiarism and AI-generated text. Results did not support any significant differences in plagiarism. However, 6% of analyzed dissertations from 2023 were positive for AI-generated text (AI scores over 20%). This is significantly different from 2013 dissertations, which only had 0% or < 20% AI scores. The findings of this study suggest that some students may be relying on generative text or editing from AI in their dissertations. Dissertation committees and awarding institutions have an obligation to ensure the students' educational and ethical development and promote academic integrity at every level of education.

**Keywords** Academic integrity · Artificial intelligence · Plagiarism · Doctoral dissertations · Misconduct · Turnitin

## Introduction

Academic misconduct is not a new issue for students, educators, or researchers (Birks & Clare, 2023; De Maio & Dixon, 2022). It is evident across grade levels and assignments – even in doctoral dissertations (Ison, 2012; McCabe, 1992). However, advances in

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technology and increases in artificial intelligence (AI) may have increased the accessibility and temptation of cheating (Cotton et al., 2024). AI tools that generate convincing, coherent texts are fairly new, such as user-friendly models like ChatGPT (Perkins, 2023); thus, quantitative research studying the utilization of AI tools in education is still developing. While AI in education is not necessarily problematic, undisclosed use or unauthorized use could be categorized as outsourcing work or unauthorized collaboration, which constitutes academic misconduct. If students excessively rely on AI to supplement their writing process without authorization to do so, especially for a capstone project such as a dissertation, they may receive undue credit for the assignment and potentially receive a degree based on work they did not do. Such unethical use of AI has serious implications for academic integrity and student assessment.

## Artificial Intelligence

While artificial intelligence (AI) technology has "existed" since the 1960s, recent advances in AI capabilities have brought AI and its implications for scholarship into mainstream discussion (Bahammam et al., 2023; Bin-Nashwan et al., 2023; Lo, 2023). Large language models (LLMs) like ChatGPT or Google's Bard can apply machine learning to large quantities of data and effectively "learn" to respond to questions and prompts and "write" new text (Birks & Clare, 2023). The breadth of training involved has enabled LLMs to imitate human language and create convincing (although, sometimes inaccurate) responses (Bahammam et al., 2023; Cotton et al., 2024). In scholarly research, AI has functioned as an author (Benichou, 2023; King, 2023), contributor (Cotton et al., 2024), analytic tool (Perkins & Roe, 2024), and research participant (Michel-Villarreal et al., 2023). While there is continued debate over AI's ability to be an author (Alser & Waisberg, 2023; Lund & Naheem, 2024), many papers have acknowledged AI contribution and many more may have used AI without acknowledgement. According to analyses in a preprint article, 1% of scholarly articles published in 2023 had evidence of LLM involvement, which corresponds to over 60,000 documents (Gray, 2024). AI is being used by some students and researchers; however, greater clarity is needed to discern how to ethically use AI in education and research (Perkins & Roe, 2024; Tang et al., 2024).

The integration of AI into higher education can innovate the learning process for students and educators, potentially increasing efficiency and expediting the research process (Benichou, 2023; Bin-Nashwan et al., 2023). Additionally, AI provides opportunities for personalization and tailoring content to meet the needs of individual students (Bahammam et al., 2023; Bond et al., 2024). However, there are also concerns about papers "plagiarizing" AI without reporting it (Alser & Waisberg, 2023; Dwivedi et al., 2023). Transparency in AI's involvement is critical to appropriately give credit where it is due (Kell et al., 2025; Tang et al., 2024). Moreover, there are other potential ethical concerns, such as, generating fake data, citing fabricated references, and perpetuating bias (Bahammam et al., 2023; Benichou, 2023; Currie, 2023). AI, while summarizing a wide span of literature, does not discriminate between the quality of the sources (i.e., Wikipedia) and may present unreliable or plagiarized information (Alser & Waisberg, 2023). AI can also hallucinate and effectively make up references and apparent "facts" (Benichou, 2023; Ji et al., 2023). If researchers and students are not careful to thoroughly analyze the text, they may unintentionally facilitate misinformation supported by nonsense references (Benichou, 2023; Ji et al., 2023). Moreover, because AI tools base text off their training data, if the source of information is biased, an overreliance on AI may inadvertently perpetuate biases (Bahammam et al.,

2023; Brandsen et al., 2024). In light of AI's ability and ease of accessibility, educators are concerned about its effect on academic integrity (Alser & Waisberg, 2023; Bahammam et al., 2023; Perkins et al., 2024).

## Academic Integrity

The degree to which faculty and students alike are committed to honesty, trust, respect, and ethical behavior in education is conceptualized as academic integrity (Bin-Nashwan et al., 2023; Holden et al., 2021). To facilitate the ethical, academic, and personal development of students, higher education institutions should encourage integrity in teaching, learning, and research (Piascik & Brazeau, 2010; Zhang et al., 2023). Unfortunately, academic misconduct is prevalent, including behaviors such as plagiarism, outsourcing assignments, cheating, unauthorized collaboration, and fabrication of results (Dorea-Bandeira et al., 2022; McCabe, 1992). Estimates of academic misconduct vary depending on the school and behavior that is being assessed. McCabe's (1992) landmark study with 6,096 college students across 31 schools reported 67% engaged in some type of academic misconduct, such as cheating on an exam, collaborating when instructed not to, copying someone else's assignment, falsifying references, or failing to cite sources. Even prestigious universities are not immune to academic misconduct. For example, 24% of students at Yale University indicated they copied someone else's answers and 14% cheated on an exam (Prihar & Wanna, 2019). Similarly, Harvard reported 23% of freshman engaged in academic misconduct (Cunningham & Shroeder, 2020). In a sample of 240 medical students, 98.3% believed themselves to be ethical; however, almost 70% reported cheating on a test (Dorea-Bandeira et al., 2022). Thus, regardless of students' perceived ethicality, grade level, or university, academic misconduct occurs.

## Academic Misconduct and Dissertations

While cheating and academic misconduct are problematic for any academic assignment, such behaviors on a doctoral dissertation are especially concerning. The dissertation process is a critical element and culmination of a doctoral degree, which requires perseverance and facilitates the academic growth of the student. Due to its challenging nature, dissertations often contribute to attrition in doctoral programs (Robinson & Tagher, 2017; Singh & Remenyi, 2018). Previous research has raised concerns that students may plagiarize or engage in contract cheating to write their doctoral dissertations (Farahian et al., 2020; Singh & Remenyi, 2018). Doctoral students who commit such misconduct in their dissertations fail to acknowledge the source of their ideas and writing, which may result in a degree based on work completed by someone else, which would not accurately reflect their competency and ability.

However, it is unclear the extent of academic misconduct in doctoral dissertations. One study analyzed the extent of plagiarism in 167 graduate research proposals using SafeAssign, which found 35.8% to 45.5% of proposals contained instances of plagiarism, varying in severity (Gilmore et al., 2010). For dissertations analyzed with Turnitin, 60% of dissertations had some level of plagiarism, as indicated by a similarity index on Turnitin above 10% (Ison, 2012). Additionally, 11% had moderate levels of plagiarism while 3% had high levels of plagiarism as defined as a similarity score over 50% (Ison, 2012). In two subsequent studies, each sample of dissertations had some dissertations with similarity scores above 50% (Ison, 2014, 2015), indicating plagiarism is a critical problem, even

in dissertations. In some cases, discovery of misconduct in dissertations has contributed to revoking the doctoral degree, dismissal of supervising faculty members, and damage to the university's reputation (Deubel, 2018). However, more recent estimates are necessary to further establish prevalence of plagiarism in doctoral dissertations.

While estimates of contract cheating are generally low, (Curtis & Clare, 2017; Curtis & Tremayne, 2021), it is unclear the extent of contract cheating for dissertations specifically. However, there are essay mills touting their ability to write dissertations, which suggests there is a potential (unethical) market for the outsourcing of dissertation work. The frequencies of contract cheating and other types of academic misconduct are elevated by their availability (De Maio & Dixon, 2022; Gilmore et al., 2010); thus, it is possible the accessibility of AI increases the temptation to outsource work, even for doctoral students (Cotton et al., 2024; Currie, 2023).

## Academic Misconduct and AI

While many have raised concerns about AI in education, there is limited research regarding students' use of AI. Many students report previously using AI technology, such as to supplement their learning or review literature (Essien et al., 2024; Segbenya et al., 2023). Students have perceived benefits to their critical thinking skills, self-efficacy, and academic motivation, although there are concerns over its accuracy and ethical use (Essien et al., 2024; Jia & Tu, 2024). Other students perceive that dependency on AI could have a negative effect on their critical thinking, reasoning, and communication skills (Segbenya et al., 2023). One self-report survey found 9.6% of Vietnamese college students reported cheating with ChatGPT (Nguyen & Goto, 2024). Moreover, when indirect questioning methods were used to reduce social desirability bias, the prevalence increased to 23.7% of students (Nguyen & Goto, 2024). However, additional research is necessary to ascertain the prevalence of AI use in higher education.

As AI provides convincing responses to a variety of questions (i.e., Benichou, 2023; Cotton et al., 2024; King, 2023; Kung et al., 2023), students may try to use AI to complete assignments or write papers and receive undue credit. Students who are transparent about AI's involvement and complete assignments in line with a professor's instructions and university policies are not committing misconduct. However, undisclosed use of AI to complete assignments is comparable to plagiarism as students take the words of someone else and present it as their own writing, without giving credit to the software that generated it (Bin-Nashwan et al., 2023; Tang et al., 2024). Unauthorized use of AI could also be conceptualized as a form of contract cheating or ghostwriting, as it reflects the submission of an assignment that was done by a third party, regardless of whether the third party was paid for the work (Bretag et al., 2020; Harper et al., 2019; Sweeney, 2023).

Doctoral students function in dual roles of being both researchers and students. As researchers, the misuse of AI may be driven by career-related motivations, such as pressure to publish, time constraints, or career advancement (Bahammam et al., 2023; Bin-Nashwan et al., 2023). As students, the lack of knowledge and skills, lack of motivation and time management, as well as fear of failure or embarrassment may drive students to plagiarize or engage in contract cheating (Ali & Alhassan, 2021). These factors intersect in motivating doctoral students to turn to AI for assistance with a dissertation. Failure to acknowledge the extent of AI's contribution to an assignment reflects a lack of transparency which can detract from the rigor and reproducibility of academic research (Kell et al., 2025). Even if students are not expressly requesting an AI tool like ChatGPT to complete

the entire assignment, an overreliance on generative suggestions may lead to outsourcing (intentional or otherwise) in which the final product is not consistent with the level of ability of the student. The prevalence of plagiarism and contract cheating indicates that some students do rely on the wording of others in their writing or even outsource assignments altogether. Moreover, these behaviors may become more accessible through AI technology. However, no known studies have evaluated the extent of AI use in doctoral dissertations.

### AI Detection

Part of the concern with AI in scholarly writing is the difficulty detecting it. As AI tools continue to advance, detection tools have attempted to keep up in a technological "arms race" (Perkins et al., 2024, p. 106). Unlike plagiarism, where a student's writing could be compared to the original source, there is no original source to which someone could compare a text that may have been generated by AI, which makes AI's involvement difficult to prove. The increasing complexity and sophistication of AI technology and the use of automatic paraphrasing technology further complicates distinguishing between AI and human writing (Bahammam et al., 2023; Weber-Wulff et al., 2023).

In a comparison of 16 AI detection tools, three tools (Turnitin, Copyleaks, and Originality.ai) demonstrated being able to accurately identify undergraduate essays versus essays written by ChatGPT 3.5 and ChatGPT-4 (Walters, 2023). Similarly, an investigation of 12 publicly available AI detection tools as well as Turnitin and PlagiarismCheck revealed that AI detectors were more accurate in identifying human written text than identifying AI-written text, especially if the document was also put through a software to paraphrase it (Weber-Wulff et al., 2023). This means the AI detectors were more likely to misclassify AI-written text as written by humans than they were to misclassify human written text as AI-written – particularly if automatic paraphrasing software was used. While Weber-Wulff and colleagues compared accuracy through several methods, Turnitin was the most accurate across all comparisons. Some AI detection tools had high rates of false positives (i.e., classifying a human-written text as AI generated), but Turnitin had zero false positives for the 54 test cases analyzed (Weber-Wulff et al., 2023). This suggests that Turnitin erred on the side of humans and any errors made were identifying AI-generated text as human-written (Weber-Wulff et al., 2023). Such a bias may be desirable for an academic setting to avoid overly penalizing a student who did not use AI. However, Turnitin is not perfect and may underestimate the percentage of AI-generated text (Perkins et al., 2024). While detection tools have limitations, they are generally more accurate than human reviewers in identifying AI-generated text (Gao et al., 2022; Perkins et al., 2024).

### Plagiarism Detection

In contrast with the challenges of AI detection, plagiarism detection has become easier with advances in technology and shifts towards open access information (Zimba & Gasparyan, 2021). Prior to the twenty-first century, detecting plagiarism was reliant on manual analysis of the text and often dependent on the subject matter knowledge of the reader and the student's writing as a whole. With increases in technology, such as the development of Turnitin and iThenticate in the late 1990s and early 2000s, wording can be compared automatically to any online sources, which enables easy identification of and comparison with possible plagiarized sources (Rahman et al., 2024). The inception of such technology was gladly received by educators and universities, to enable swift identification of plagiarism

and mitigate misconduct (Batane, 2010; Rahman et al., 2024). Subsequently, tools like Turnitin and iThenticate became widely used across academia (Heckler et al., 2013; Rahman et al., 2024).

Software to compare texts may also encourage students to more actively avoid plagiarism, as they perceive any plagiarism will be caught through such software (Batane, 2010; Ison, 2015). However, Batane (2010) notes that Turnitin detection software is still limited and may overly flag texts as plagiarized. For example, several repeated words – even innocuous words, like "to the" or "et al." – may be flagged as similar to another text. Thus, plagiarism detection is still contingent on personal evaluations and judgements of the professor; however, the software enables quick comparison with a flagged source, informing the professor's decision of whether plagiarism occurred (Batane, 2010; Rahman et al., 2024; Zimba & Gasparyan, 2021).

Even if misconduct is undetected, such unethical behavior has deleterious effects. For the student who cheats, they cheat themselves of an opportunity for intellectual growth by passively engaging in educational tasks and only developing surface-level understanding, to the detriment of their critical thinking and ability (Schei et al., 2024; Segbenya et al., 2023; Sweeney, 2023). Across academic levels, misconduct detracts from the rigor and credibility of legitimate research and assessment (Elali & Rachid, 2023; Liverpool, 2023; Perkins et al., 2024; Sweeney, 2023), reflecting poorly on the university. Furthermore, awarding a degree carries the expectation of certain knowledge or skills (Amigud & Lancaster, 2019). If misconduct was used to achieve the degree, such unethical behavior may have cascading effects on the workplace (Sweeney, 2023). As plagiarism and outsourcing have been identified in doctoral dissertations (Ison, 2012, 2014), doctoral students may be similarly susceptible to the temptation of AI for outsourcing work. Thus, the goal of the present study is to address a significant gap in the literature regarding the scope of academic misconduct in dissertations and evaluate the prevalence of AI usage and plagiarism in doctoral dissertations.

## The Current Study

While there is a long history of research on academic misconduct (Gallant & Rettinger, 2022; McCabe, 1992), recent advances in AI may increase accessibility for cheating behavior, such as unauthorized outsourcing of assignments. Considering the challenges and attrition associated with doctoral dissertations (Young et al., 2019), doctoral students may be tempted to use AI to supplement their writing process, such as to synthesize literature, paraphrase articles, or simply overcome writer's block (Bin-Nashwan et al., 2023; Segbenya et al., 2023). While several studies have indicated plagiarism is a concern, even in doctoral dissertations (Ison, 2014, 2015), no known studies have investigated the extent of AI usage in doctoral dissertations. Consequently, the goal of the present study is to conduct a comparative analysis exploring AI prevalence in doctoral dissertations using Turnitin. Specifically, we will compare dissertations from 2013, completed before widespread availability of convincing generative AI with dissertations from 2023, completed after widespread availability of generative AI. Moreover, plagiarism scores will also be compared to enable thorough consideration of the prevalence of academic misconduct in dissertations.

## Data Collection

To inform the collection of dissertations used for the present analysis, we first conducted an a priori power analysis with G*Power to determine the necessary sample size. Considering prior studies have indicated a significant skew for similarity scores in dissertations (Ison, 2014), we intended to use non-parametric analyses. To detect a medium effect size (Cohen's $d = 0.50$ with 0.90 power in a two-tailed Mann-Whiteney $U$ test ($\alpha = 0.05$), G*Power recommended a minimum sample size of 180. Subsequently, 200 dissertations were collected to conduct the desired analyses. To collect the dissertations, we searched the EBSCOhost Open Dissertations database for dissertations posted within certain time frames. This database was selected as it provides free access to full texts for millions of dissertations and theses (EBSCO, 2024). For the purposes of this study, dissertations posted between January 2013 and December 2013 comprised the dissertations written prior to the widespread availability and accessibility of AI (Walters, 2023). Of the initial 49,678 results, dissertations were randomly selected and evaluated for eligibility until the required sample size ($n = 100$) was achieved. Similarly, dissertations posted between January 2023 and December 2023 were randomly selected to reflect dissertations written after widespread availability and accessibility of AI. Of the initial 38,789 dissertations in 2023, randomly selected texts were evaluated for eligibility until sample requirements were met ($n = 100$). Dissertations were ultimately eligible to be selected for the study provided their full text was openly accessible, the text was in English, it was a doctoral dissertation (as opposed to a master's level dissertation or thesis), and the awarding university is accredited in the United States or Canada, which is comparable to the selection criteria applied in previous research (i.e., Ison, 2015).

## Analysis Plan

Dissertations that met the inclusion criteria were downloaded by the researchers and scanned through Turnitin software so the plagiarism and AI reports could be reviewed. To produce an AI score, the document submitted to Turnitin could not exceed 30,000 words. For particularly lengthy dissertations, the document was divided into equal parts and analyzed separately to meet word count limits. Subsequently, an average of the scores for the individual parts was computed to provide an overall score for the document as a whole. Turnitin was selected for use because of its frequent utilization in higher education (Arabyat et al., 2022), effectiveness in identifying plagiarism (Rahman et al., 2024), and success in distinguishing AI from human-generated text (Weber-Wulff et al., 2023). Additionally, Turnitin has been used in previous publications for comparative analyses (Ison, 2014, 2015). References, acknowledgements, title pages, table of contents, and definitions of abbreviations were removed from documents prior to running plagiarism analyses (Ison, 2014, 2015). For the plagiarism report, similarity sources were excluded if the flagged text was quoted, identified as corresponding to the original dissertation, or a subsequent journal publication written by the original author (i.e., a publication directly derived from their dissertation). Additionally, flagged text was excluded if the matched source was published after the dissertation acceptance date. Descriptive statistics were also collected for the dissertations including geographic region of the awarding university, academic discipline, and length of document.

The percentage of plagiarism and AI were input into SPSS (v.28.0). Considering previous research has indicated plagiarism scores are significantly skewed and not normally distributed (Ison, 2014, 2015), we anticipated using non-parametric analyses for comparison. Specifically, Mann–Whitney *U* tests were used to compare the AI and plagiarism scores for dissertations posted in 2023 with dissertations posted in 2013. Following methods for comparative analyses in the literature (Ison, 2014, 2015), we categorized dissertations based on the level of plagiarism present: no plagiarism (< 10%), low (11–24%), moderate (25–49%) and high (> 50%).

To our knowledge, there are no currently established categorizations of AI generation scores; however, Turnitin does not produce an exact percentage for scores under 20%, because they are cautioned as being inaccurate. For quantitative analyses, scores classified as "under 20%" were treated as 10%. However, there can be varying amounts in the under 20% category because AI scores were often averaged across several subparts of individual dissertations to meet word count limits. Thus, if a document has a score of < 20% on one part and a score of 0% on a second part, their final AI score is counted as less than someone who has a < 20% score on both parts of the dissertation. Considering planned analyses were non-parametric, which function through ranking the data, this approach enables scores in the < 20% range to be ranked in a manner consistent with their estimated AI use. Subsequently, we categorized dissertations based off AI generation scores into following the categories: none (0%), very low (< 20%), low (20–35%), moderate (35–50%), and high (> 50%). The AI and plagiarism categorizations of the two groups were then compared with chi-square tests for independence.

## Results

The majority of dissertations were from public institutions (88.0%) in the Southern region of the United States (50.5%). The dissertations were primarily for PHD degrees (87.5%) in the natural science (22.5%) or social science (19.0%) disciplines (Table 1).

### Preliminary Analysis

An independent samples *t*-test between the two groups indicated there was not a significant difference between dissertations published in 2013 and 2023 in the length of the document or total length scanned after removing extraneous material, such as title page, table of contents, and appendices (Table 2).

Consistent with past research (Ison, 2015), the distribution of plagiarism scores was significantly skewed and the Shapiro–Wilk statistic was significant for both AI and plagiarism percentages from both 2013 and 2023 ($p < 0.001$). In line with expectations, the normal Q-Q plot for AI scores indicated substantial deviation from normality, which supported the planned non-parametric analyses. Figures 1 and 2 contain the frequency histograms for AI and plagiarism percentages. A nonparametric correlation was conducted with Spearman's rho, which indicated AI scores weakly positively correlated with plagiarism scores ($\rho = 0.171$, $p = 0.015$). Additionally, total length of the document negatively correlated with both AI scores ($\rho = -0.144$, $p = 0.042$) and plagiarism scores ($\rho = -0.177$, $p = 0.012$) indicating that shorter documents were associated with higher AI and plagiarism estimates. However, the correlations had different patterns of significance when separated by year

**Table 1** Dissertation Institution and Degree Descriptions

| | Total $N=200$ | 2013 $N=100$ | 2023 $N=100$ |
|---|---|---|---|
| Institution Type | | | |
| Public | 166 (88.0%) | 88 | 78 |
| Private | 34 (17.0%) | 12 | 22 |
| Geographical Region | | | |
| Northeast U.S | 18 (9.0%) | 9 | 9 |
| Midwest U.S | 44 (22.0%) | 21 | 23 |
| South U.S | 101 (50.5%) | 56 | 45 |
| West U.S | 31 (15.5%) | 11 | 20 |
| Canada | 6 (3.0%) | 3 | 3 |
| Discipline | | | |
| Business | 10 (5.0%) | 3 | 7 |
| Communication | 4 (2.0%) | 3 | 1 |
| Computer Science | 10 (5.0%) | 4 | 6 |
| Education | 32 (16.0%) | 13 | 19 |
| Engineering | 31 (15.5%) | 11 | 20 |
| Humanities | 14 (7.0%) | 8 | 6 |
| Mathematics | 7 (3.5%) | 5 | 2 |
| Medicine | 9 (4.5%) | 5 | 4 |
| Natural Science | 45 (22.5%) | 25 | 29 |
| Social Science | 38 (19.0%) | 23 | 15 |
| Degree Type | | | |
| Doctor of Philosophy (PHD) | 175 (87.5%) | 93 | 82 |
| Doctor of Business Administration (DBA) | 1 (0.5%) | 0 | 1 |
| Doctor of Healthcare Administration (DHA) | 2 (1.0%) | 0 | 2 |
| Doctor of Musical Arts (DMA) | 1 (0.5%) | 0 | 1 |
| Doctor of Ministry (DMin) | 1 (0.5%) | 0 | 1 |
| Doctor of Education (EDD) | 17 (8.5%) | 6 | 11 |
| Doctor of Professional Studies (DPS) | 1 (0.5%) | 0 | 1 |
| Doctor of Psychology (PsyD) | 2 (1.0%) | 1 | 1 |

Because the samples for 2013 and 2023 are both 100, the reported number in their respective columns represents both the frequency and the percentage

**Table 2** Dissertation Page Length

| | 2013 | | 2023 | | Independent *t*-test | | |
|---|---|---|---|---|---|---|---|
| | *M* (*SD*) | Range | *M* (*SD*) | Range | *t*(198) | *p* | *d* |
| Total length | 194.22 (106.88) | 53–699 | 176.52 (78.79) | 68–496 | 1.333 | 0.184 | 0.19 |
| Total length scanned | 128.37 (68.12) | 29–419 | 126.07 (61.06) | 36–354 | 0.251 | 0.802 | 0.04 |

Total length scanned refers to the scannable pages of the document after removing reference material, title page, table of contents, and appendices

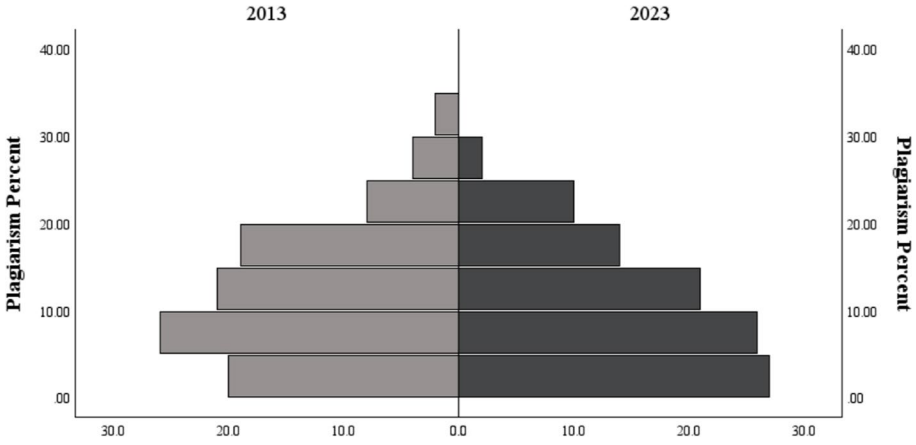
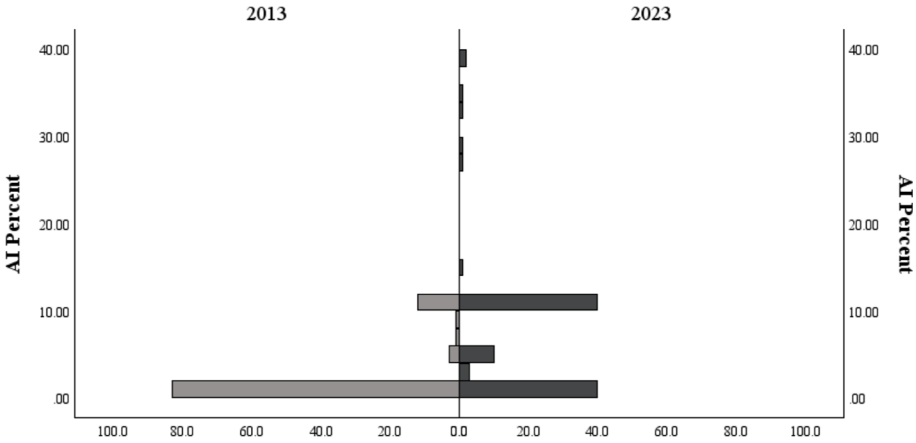



**Fig. 1** Frequency Histogram of Plagiarism Scores by Year




*Note*. Turnitin does not produce estimates for scores under 20%, rather, they are given a score of "*%" as scores that low are considered to be within an error range. To be treated numerically, "*%" scores were recoded as 10%. For dissertations that were analyzed in separate portions to meet word count limits, the AI scores were averaged together. Thus, someone with a score of "*%" on two dissertation portions would have an overall score of 10%; whereas, someone with a score of "*%" on one portion and 0% on another portion would have an overall score of 5%. As the Mann-Whitney U test is non-parametric and operates by ranking the data, coding "less than 20%" as "10%" should not affect the rank of the scores.

**Fig. 2** Frequency Histograms of AI Percent Scores by Year

(See Table 3). Notably, the positive correlation between AI scores and plagiarism scores was stronger in the 2023 group and insignificant in the 2013 group.

## Plagiarism

An independent-samples Mann–Whitney *U* test was conducted to compare the distribution of plagiarism scores between dissertations published in 2013 and dissertations published in 2023. The results did not find a significant difference between groups ($U = 4549.50$,

**Table 3** Nonparametric Correlation Coefficients

| | Total length scanned | AI | Plagiarism |
|---|---|---|---|
| Total length scanned | - | −0.241* | −0.104 |
| AI Percentage | −0.055 | - | 0.291** |
| Plagiarism Percentage | −0.240* | 0.177 | - |

Lower diagonal reports 2013 dissertations while upper diagonal reports 2023 dissertations

*$p<0.05$; **$p<0.01$

$z=-1.10$, $p=0.271$). Dissertations published in 2023 (Mdn = 9.60) were not significantly different in plagiarism scores than dissertations published in 2013 (Mdn = 10.01). Subsequently, dissertations were categorized as no plagiarism (< 10%), low plagiarism (11–24%), moderate (25–49%), and high (50%). Results indicate 49% of dissertations completed in 2013 had some plagiarism (> 10% on Turnitin) while 44% of dissertations completed in 2023 had some plagiarism (See Table 4). An independent chi-square analysis of frequency of dissertations across categories indicated there was not a significant difference between 2013 and 2023 dissertations ($\chi^2=3.11$, $p=0.211$, Cramer's V = 0.125).

## Artificial Intelligence

An independent-samples Mann–Whitney *U* test was conducted on the dissertations to compare the distribution of AI scores between dissertations published in 2013 and dissertations published in 2023. Dissertations with a score of "less than 20%" were treated as 10% for the analyses to provide an analyzable number, which can be ranked accordingly. The results indicate a significant difference between groups ($U=7173.00$, $z=6.09$, $p<0.001$). Dissertations published in 2023 (Mdn = 5.12) had significantly higher AI scores than dissertations published in 2013 (Mdn = 0.00). Subsequently, dissertations were categorized into no AI (0%), very low (< 20%), low (20–35%), moderate (35–50%), and high (50 +), which are summarized in Table 5. The highest AI score fell in the less than 20% range for 2013 dissertations, while 2023 dissertations had scores as high as 39%. Comparison of the groups with a chi-square test for independence further supported significant differences between dissertations published in 2013 and dissertations published in 2023 ($\chi^2=36.67$,

**Table 4** Plagiarism Categories by Year

| | No plagiarism (< 10%) | Low (11–24%) | Moderate (25–50%) | High (50% +) | Total plagiarism |
|---|---|---|---|---|---|
| 2013 | 50% | 42% | 7% | 0% | 49% |
| 2023 | 56% | 42% | 2% | 0% | 44% |

**Table 5** Categorical Distribution of AI Scores

| | None (0%)* | Very low (< 20%)* | Low (20–35%)* | Moderate (35–50%) | High (50% +) |
|---|---|---|---|---|---|
| 2013 | 81% | 19% | 0% | 0% | 0% |
| 2023 | 40% | 54% | 4% | 2% | 0% |

*significant difference between 2013 and 2023 in proportion of dissertations in that category, with Bonferroni correction

$p < 0.001$, Cramer's $V = 0.428$). None of the dissertations that were positive for AI (scores above 20%) indicated any acknowledgement of AI use in the text.

## Discussion

The purpose of this study was to explore the prevalence of plagiarism and artificial intelligence (AI) use in doctoral dissertations. While plagiarism is clearly categorized as academic misconduct, the use of AI is less clear. AI use can be ethical, provided it is acknowledged, and its use aligns with assignment instructions, class guidelines, and institutional policy. However, utilizing AI in violation of explicit guidelines would constitute an unauthorized tool, which would be cheating (Nguyen & Goto, 2024). Moreover, utilizing AI without acknowledgement is arguably a misrepresentation of someone—or something—else's words as one's own without citation, which is analogous to plagiarism or contract cheating (Dwivedi et al., 2023; Tang et al., 2024). Consequently, the discussion of the results is framed by the view of plagiarism and unacknowledged AI use as indications of misconduct.

### Plagiarism

Results of this study found that almost half of doctoral dissertations from both 2013 and 2023 had some level of plagiarism, indicated by a Turnitin score of over 10%. This is comparable to estimates of plagiarism previously observed in college students (Curtis & Tremayne, 2021), graduate research proposals (Gilmore et al., 2010) and doctoral dissertations (Ison, 2012, 2015). Additionally, comparison of plagiarism scores indicates there was not a significant difference between dissertations in 2013 and 2023 in the prevalence or severity of plagiarism. While some research has suggested plagiarism decreased from the 1990s to 2015 (Ison, 2015), the current study did not observe a continuation of that trend. However, Ison (2015) reviewed dissertations that were 20 years apart in contrast with the dissertations being 10 years apart in the present study, which may have been a factor in the discrepant results.

Alternatively, the widespread utilization of plagiarism detection software in the early 2000s (Batane, 2010; Rahman et al., 2024), could have played a role in more stable plagiarism rates. Concerns about misconduct have elevated in the educational sector with the advent of generative AI (Bahammam et al., 2023); however, concerns about plagiarism escalated prior to 2013, with advances in the internet enabling easier copying-and-pasting (Bailey, 2019; Batane, 2010; Ison, 2015). Consequently, institutional emphases on avoiding plagiarism and utilizing plagiarism detection software may have been generally consistent over the time from 2013 to 2023, contributing to the insignificant difference in the prevalence of plagiarism.

The consistently common occurrence of plagiarism draws attention to the critical necessity of institutional efforts to encourage academic integrity, even for doctoral candidates (Gilmore et al., 2010; Ison, 2015). Across academic levels, students may be tempted to plagiarize if they feel overwhelmed by academic pressure or limited by time constraints. Additionally, lack of knowledge on how to effectively avoid plagiarism can contribute to unintentional plagiarism (Ali & Alhassan, 2021). Consequently, students must be equipped to competently engage with research and paraphrase and quote effectively, with appropriate citations. Moreover, the dissertation committee should recognize their responsibility in the academic rigor and integrity of dissertations (Deubel, 2018). In addition to the fundamental

knowledge and practical application for specific disciplines, doctoral programs should seek to foster ethics as a professional skill in their students.

## Artificial Intelligence

The importance of ethical competencies integrated into educational curricula is particularly relevant for AI. Many students are familiar with AI applications and report using them in their studies to facilitate their understanding and condense literature (Essien et al., 2024; Segbenya et al., 2023); moreover, estimates of cheating with AI range from roughly 10 to 24% (Nguyen & Goto, 2024). This study adds to empirical assessments of AI use, suggesting even doctoral dissertations are not immune, with 6% of dissertations flagged as including text generated by AI (Turnitin score > 20%). AI use is not necessarily unethical, pointing to the necessity of fostering ethical responsibility so students can critically evaluate their use of AI. Viewing the use of AI without permission or acknowledgement as an unauthorized outsourcing of academic work, the prevalence of positive scores points to the possible use and misuse of AI in doctoral dissertations. Moreover, two dissertations had 39% of their document flagged, which indicates some students may rely on the wording of others to complete portions of the assignment.

None of the documents that were flagged for AI reported utilizing any AI. At the very least, this suggests that students need to acknowledge their use of AI and be transparent in the development of their dissertation. At worst, it could indicate unethical outsourcing in which students are relying on AI or generative suggestions while drafting their dissertation in violation of academic or committee guidelines. Considering the dissertation process should challenge students to develop critical reasoning and higher-order thinking, as well as demonstrate their competency and ability to meaningfully contribute to the field, overreliance on AI may provide a cognitive shortcut that reduces the student's personal growth and ability to deeply engage in the research. Previous research has also suggested that Turnitin errs on the side of students (Weber-Wulff et al., 2023), which raises the concern that actual use of AI may be higher, especially if automatic paraphrasing tools are also incorporated (Weber-Wulff et al., 2023).

For dissertations completed in 2013, prior to increased functionality and accessibility of AI, any positive scores should be considered false positives. Scores above 0 occurred on 19% of dissertations completed in 2013; however, they were all flagged as less than 20%, which Turnitin no longer considers a "positive" result for AI (Turnitin, 2024). This bolsters the perception that the range of scores from 1 to 19% is a margin of error that does not necessarily indicate AI usage. Operating with the definition of a positive score for AI as 20% or higher, there were zero false positives for the 2013 dissertations. However, it is critical to note that failure to find false positives in this study does not mean false positives are not possible. Consequently, if assignments are flagged as positive for AI with a score above 20%, instructors should review the flagged text carefully within the context of the student's submission and previous work to determine whether AI use has occurred and whether it violates assignment instructions or university-level policies.

## Implications for Educators

The findings of this study point to critical implications for higher education, particularly for doctoral dissertations. AI's involvement is difficult to prove or disprove; consequently, developing AI detection in conjunction with advances in AI technology is critical to ensure

unethical behavior is not missed and ethical behavior is not unduly penalized. Collaboration between AI developers and AI detectors can support efforts to maintain academic and research integrity. Considering other tools, such as Crossplag, GPT Zero, and SEO.ai have previously produced false positives (Walters, 2023; Weber-Wulff et al., 2023), it is critical for educators to critically evaluate the evidence supporting the tool before arbitrarily applying it to students' submissions.

Dissertation drafts should be analyzed through an evidence-based tool, such as Turnitin (Weber-Wulff et al., 2023), throughout the dissertation process. While there are limitations to such tools, the resulting analysis can provide insight into the document and provide additional information for the dissertation committee to weigh in deciding whether misconduct has occurred. Considering the maximum word count for Turnitin to generate an estimate of AI usage is currently 30,000 words, educators should consider whether they need to analyze portions of dissertation drafts to ensure that an AI estimate is provided. Additionally, early review and active involvement of the dissertation chair and committee in the dissertation process is critical to provide support for the student. This can enable problems to be identified and corrected, providing students with the opportunity to learn from their mistakes and remediate unethical behavior, further promoting ethical competency as a learnable skill.

It is also critical to thoroughly evaluate instances of plagiarism to determine the degree of similarity between the student's text and the highlighted source. While documents with less than 25% plagiarism are highlighted as green in Turnitin, for lengthy documents, such as dissertations, that amount may still be problematic. Educators should not apply arbitrary cut-offs for what constitutes an acceptable Turnitin plagiarism score; rather, they should make an informed decision, contextualized by their own familiarity with the discipline and its terminology to ascertain whether a student's text is sufficiently paraphrased and reflects the student's original work. In addition to plagiarism scores, dissertations and other assignments should be evaluated for errors in references and citations, which can be substantial in student research (Moore, 2014), to ensure students are appropriately citing original material.

Educators should also ensure there are clear expectations for AI use in assignments. For some assignments, it is warranted to prohibit AI use; however, educators should consider how they can promote critical use of AI, leveraging its benefits while fostering AI literacy, critical thinking, and creative application (Bond et al., 2024; Essien et al., 2024; Jia & Tu, 2024). While AI increases the accessibility of misconduct, which can elevate the likelihood of such behaviors (De Maio & Dixon, 2022), AI also provides a valuable tool that can be utilized collaboratively to facilitate learning (Jia & Tu, 2024; Luo, 2024). Luo (2024) argues for viewing AI involvement as a continuum reflecting different collaborative contributions from the student and from AI, which highlights the importance of a nuanced view of AI and its implications for ethical conduct in academia.

For dissertation committees and institutions that adopt a more open stance towards AI in the dissertation process, it is still important to foster ethical use of AI, for example, requiring students to detail the scope and role of AI in their submission (Tang et al., 2024). Moreover, considering the possible limitations of AI with hallucinations, plagiarized wording, bias, and unreliable sources (Alser & Waisberg, 2023; Bahammam et al., 2023; Ji et al., 2023), students who use AI should articulate steps taken to ensure rigor and quality for any AI-generated contributions. In addition to promoting integrity, being open and transparent about methodology and technology utilized through

the research process is critical for reproducible, rigorous research (Kell et al., 2025). However, future research is needed to identify best practices for rigorous use of AI in research and education.

Considering the challenges of plagiarism and misconduct more broadly, university- and program-level efforts are critical to facilitate a climate of academic integrity that can encourage student's educational and ethical growth (Packalen & Rowbotham, 2024). Furthermore, higher education institutions must have clarity in policy and provide continual information and support for faculty and students as AI advancements shape a new academic landscape (Nguyen, 2025). Considering there is substantial variation in policies regarding AI usage in academia (Moorhouse et al., 2023), it is critical for the development of clear guidelines as to what constitutes ethical use of AI. The implementation of AI technology and its impact on the originality of a student's work is necessarily context dependent, situated within the expectations and learning outcomes for a specific assignment (Luo, 2024); consequently, a collaborative approach between policymakers, educators, and students would be beneficial when weighing guidelines for use in light of the goals of education.

## Limitations and Directions for Future Research

While this study collected a random sample of dissertations from accredited institutions across the United States and Canada, future research is needed to replicate these effects in a larger sample of dissertations. The dissertations were primarily public institutions, and in science or education disciplines, which limits the generalizability of the findings to private institutions and other disciplines, such as mathematics, communication, and business. Additionally, due to the archival nature of the collection process, it was beyond the scope of this study to ascertain whether the potential use of artificial intelligence was prohibited or permitted by the dissertation committee or institution. It is possible that students were allowed to use AI-tools, such as ChatGPT, to aid in the writing process. This study underscores the broader need of ascertaining the prevalence of AI use and misuse in higher education.

This study is also limited by technology's ability to provide an accurate estimate of AI usage. While plagiarized text can be compared with an original source, no such source exists for AI-generated text, which makes it difficult to definitively know whether the text was generated by AI. Several studies have illustrated the accuracy of Turnitin; however, it is possible for text to be misidentified (Perkins et al., 2024; Weber-Wulff et al., 2023). Consequently, a positive AI score, even through Turnitin, is not conclusive proof of AI's involvement. Conversely, a 0% AI score does not prove AI was not involved, especially considering the prevalence of paraphrasing tools or other strategies for reworking AI-generated text (Weber-Wulff et al., 2023). Inclusion of the 2013 dissertations as a control group helps bolster the estimate of AI usage as above the margin of error for misidentification, but it is still important to note the challenges of quantifying AI usage. Moreover, Turnitin does not produce exact estimates for scores under 20%. While scores within that range were treated numerically as "10%" this may not accurately reflect the variation of possible scores under 20%. While ranking the data through nonparametric analysis should help mitigate that, future research could consider exploring the prevalence of AI as reported by additional AI detection tools with converging evidence from self-report surveys.

## Conclusion

In conclusion, this study presents empirical evidence that AI usage is occurring, even in assignments as prestigious as doctoral dissertations. Moreover, plagiarism remains a consistent problem in higher education. The dissertation should reflect the level of ability of the student to critically engage with research and advance their field. Thus, through clear university policies, expansion in program-level training on ethics and misconduct, and continual mentorship from the dissertation committee, doctoral students can be equipped with the critical thinking, content knowledge, and ethical application for the dissertation journey. It is evident that doctoral students, who are developing their theoretical knowledge as students and practical applications skills as researchers, must also cultivate their academic integrity. An emphasis on integrity at every level of academia can facilitate individual-level ethics, bolster programmatic rigor, and ensure the credibility of research and assessment, ultimately fostering an ethical climate for learning and research.

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## Declarations

**Research Involving Humans and Animals Statement** This research is exempt from IRB review because it does not involve human subjects.

**Conflicts of interest** The authors have no known conflicts of interest to disclose.

## References

Ali, H. I., & Alhassan, A. (2021). Fighting contract cheating and ghostwriting in higher education: Moving towards a multidimensional approach. *Cogent Education, 8*(1), 1–18. https://doi.org/10.1080/2331186X.2021.1885837

Alser, M., & Waisberg, E. (2023). Concerns with the usage of ChatGPT in academia and medicine: A viewpoint. *American Journal of Medicine Open, 9*(1), 1–2. https://doi.org/10.1016/j.ajmo.2023.100036

Amigud, A., & Lancaster, T. (2019). 246 reasons to cheat: An analysis of students' reasons for seeking to outsource academic work. *Computers & Education, 134*(1), 98–107. https://doi.org/10.1016/j.compedu.2019.01.017

Arabyat, R. M., Qawasmeh, B. R., Al-Azzam, S. I., Nusair, M. B., & Alzoubi, K. H. (2022). Faculty members' perceptions and attitudes towards anti-plagiarism detection tools: Applying the theory of planned behavior. *Journal of Empirical Research on Human Research Ethics, 17*(3), 275–283. https://doi.org/10.1177/15562646221078655

Bahammam, A. S., Trabelsi, K., Pandi-Perumal, S. R., & Jahrami, H. (2023). Adapting to the impact of artificial intelligence in scientific writing: Balancing benefits and drawbacks while developing policies and regulations. *Journal of Nature and Science of Medicine, 6*(3), 152–158. https://doi.org/10.4103/jnsm.jnsm_89_23

Bailey, J. (2019). 5 historical moments that shaped plagiarism: Plagiarism since the dawn of language. *Turnitin*. https://www.turnitin.com/blog/5-historical-moments-that-shaped-plagiarism. Accessed 3 Apr 2025.

Batane, T. (2010). Turning to Turnitin to fight plagiarism among university students. *Educational Teaching & Society, 13*(2), 1–12.

Benichou, L., ChatGPT. (2023). The role of using ChatGPT AI in writing medical scientific articles. *Journal of Stomatology, Oral and Maxillofacial Surgery, 124*(5), 1–3. https://doi.org/10.1016/j.jormas.2023.101456

Bin-Nashwan, S. A., Sadallah, M., & Bouteraa, M. (2023). Use of ChatGPT in academia: Academic integrity hangs in the balance. *Technology in Society, 75*(1), 1–11. https://doi.org/10.1016/j.techsoc.2023.102370

Birks, D., & Clare, J. (2023). Linking artificial intelligence facilitated academic misconduct to existing prevention frameworks. *International Journal for Educational Integrity, 19*(20), 1–10. https://doi.org/10.1007/s40979-023-00142-3

Bond, M., Khosravi, H., Laat, M. D., Bergdahl, N., Negrea, V., Oxley, E., Pham, P., Chong, S. W., & Siemens, G. (2024). A meta systematic review of artificial intelligence in higher education: A call for increased ethics, collaboration, and rigour. *International Journal of Educational Technology in Higher Education, 21*(4), 1–41. https://doi.org/10.1186/s41239-023-00436-z

Brandsen, S., Chandrasekhar, T., Franz, L., Grapel, J., Dawson, G., & Carlson, D. (2024). Prevalence of bias against neurodivergence-related terms in artificial intelligence language models. *Autism Research, 17*(2), 234–248. https://doi.org/10.1002/aur.3094

Bretag, T., Harper, R., Rundle, K., Newton, P. M., Ellis, C., Sonia, Saddiqui, S., & van Haeringen, K. (2020). Contract cheating in Australian higher education: A comparison of non-university higher education providers and universities. *Assessment & Evaluation in Higher Education, 45*(1), 125–139. https://doi.org/10.1080/02602938.2019.1614146

Cotton, D. R. E., Cotton, P. A., & Shipway, J. R. (2024). Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. *Innovations in Education and Teaching International, 61*(2), 228–239. https://doi.org/10.1080/14703297.2023.2190148

Cunningham, J. P., & Shroeder, L. F. (2020). Class of 2019 by the numbers. *The Harvard Crimson*. https://features.thecrimson.com/2015/freshman-survey/academics-narrative/. Accessed 8 May 2024.

Currie, G. M. (2023). Academic integrity and artificial intelligence: Is ChatGPT hype, hero or heresy? *Seminars in Nuclear Medicine, 53*(5), 719–730. https://doi.org/10.1053/j.semnuclmed.2023.04.008

Curtis, G. J., & Clare, J. (2017). How prevalent is contract cheating and to what extent are students repeat offenders? *Journal of Academic Ethics, 15*(2), 115–124. https://doi.org/10.1007/s10805-017-9278-x

Curtis, G. J., & Tremayne, K. (2021). Is plagiarism really on the rise? Results from four 5-yearly surveys. *Studies in Higher Education, 46*(9), 1816–1826. https://doi.org/10.1080/03075079.2019.1707792

De Maio, C., & Dixon, K. (2022). Promoting academic integrity in institutions of higher learning: What 30 years of research (1990–2020) in Australasia has taught us. *Journal of College and Character, 23*(1), 6–20. https://doi.org/10.1080/2194587X.2021.2017972

Deubel, P. (2018). Punishment or policy change: A case of plagiarism in a dissertations. *Journal of Educational Research and Practice, 8*(1), 101–112. https://doi.org/10.5590/JERAP.2018.08.1.08

Dorea-Bandeira, I., Oliveira, V. G., Araujo-Pereira, M., Andrade, B. B., Nazar, A. N., Quintanilha, L. F., & Avena, K. M. (2022). Academic misconduct: Attitudes and perceptions of Brazilian medical students. *Problems of Education in the 21 Century, 80*(3), 426–437. https://doi.org/10.33225/pec/22.80.426

Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., Baabdullah, A. M., Koohang, A., Raghavan, V., Ahuja, M., Albanna, H., Albashrawi, M. A., Al-Busaidi, A. S., Balakrishnan, J., Barlette, Y., Basu, S., Bose, I., Brooks, L., Buhalis, D., … & Wright, R. (2023). "So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management*, *71*(1), 1–63. https://doi.org/10.1016/j.ijinfomgt.2023.102642

EBSCO. (2024). *EBSCO open dissertations*. https://www.ebsco.com/products/research-databases/ebsco-open-dissertations. Accessed 11 June 2024.

Elali, F. R., & Rachid, L. N. (2023). AI-generated research paper fabrication and plagiarism in the scientific community. *Patterns, 4*(3), 1–4. https://doi.org/10.1016/j.patter.2023.100706

Essien, A., Bukoye, O. T., O'Dea, X., & Kremantzis, M. (2024). The influence of AI text generators on critical thinking skills in UK business schools. *Studies in Higher Education, 49*(5), 865–82. https://doi.org/10.1080/03075079.2024.2316881

Farahian, M., Parhamnia, F., & Avarzamani, F. (2020). Plagiarism in theses: A nationwide concern from the perspective of university instructors. *Cogent Social Sciences, 6*(1), 1–17. https://doi.org/10.1080/23311886.2020.1751532

Gallant, T. B., & Rettinger, D. (2022). An introduction to 30 years of research on academic integrity. *Journal of College and Character, 23*(1), 1–5. https://doi.org/10.1080/2194587X.2021.2017975

Gao, C. A., Howard, F. M., Markov, N. S., Dyer, E. C., Ramesh, S., Luo, Y., & Pearson, A. T. (2022). Comparing scientific abstracts generated by ChatGPT to original abstracts using an artificial intelligence output

detector, plagiarism detector, and blinded human reviewers. *Nature Portfolio Journal: Digital Medicine, 6*(75), 1–5. https://doi.org/10.1101/2022.12.23.521610

Gilmore, J., Strickland, D., Timmerman, B., Maher, M., & Feldon, D. (2010). Weeds in the flower garden: An exploration of plagiarism in graduate students' research proposals and its connection to enculturation, ESL, and contextual factors. *International Journal for Educational Integrity, 6*(1), 1–17. https://doi.org/10.21913/IJEI.v6i1.673

Gray, A. (2024). *ChatGPT "contamination": Estimating the prevalence of LLMs in the scholarly literature* [Preprint]. Cornell University. https://arxiv.org/abs/2403.16887. Accessed 3 May 2024.

Harper, R., Bretag, T., Ellis, C., Newton, P., Rozenberg, P., Saddiqui, S., & van Haeringen, K. (2019). Contract cheating: A survey of Australian university staff. *Studies in Higher Education, 44*(11), 1857–1873. https://doi.org/10.1080/03075079.2018.1462789

Heckler, N. C., Rice, M., & Bryan, C. H. (2013). Turnitin systems: A deterrent to plagiarism in college classrooms. *Journal of Research on Technology in Education, 453*(3), 229–248. https://doi.org/10.1080/15391523.2013.10782604

Holden, O. L., Norris, M. E., & Kuhlmeier, V. A. (2021). Academic integrity in online assessment: A research review. *Frontiers in Education, 6*(1), 1–13. https://doi.org/10.3389/feduc.2021.639814

Ison, D. C. (2012). Plagiarism among dissertations: Prevalence at online institutions. *Journal of Academic Ethics, 10*(3), 1–13. https://doi.org/10.1007/s10805-012-9165-4

Ison, D. C. (2015). The influence of the internet on plagiarism among doctoral dissertations: An empirical study. *Journal of Academic Ethics, 13*(2), 151–166. https://doi.org/10.1007/s10805-015-9233-7

Ison, D. C. (2014). Does the online environment promote plagiarism? A comparative study of dissertations from brick-and-mortar versus online institutions. *Journal of Online Learning and Teaching*, *10*(2), 272–283. https://commons.erau.edu/ww-graduate-studies/4/. Accessed 28 Apr 2024.

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys, 55*(12), 1–38. https://doi.org/10.1145/3571730

Jia, X.-H., & Tu, J.-C. (2024). Towards a new conceptual model of AI-enhanced learning for college students: The roles of artificial intelligence capabilities, general self-efficacy, learning motivation, and critical thinking awareness. *Systems, 12*(3), 74. https://doi.org/10.3390/systems12030074

Kell, C., Kraeer, B., & Cain, W. (2025). Can AI facilitate a human-centric approach to writing a problem of practice dissertation? *Impacting Education, 10*(1), 1–5. https://doi.org/10.5195/ie.2025.482

King, M. R., ChatGPT. (2023). A conversation on artificial intelligence, chatbots, and plagiarism in higher education. *Cellular and Molecular Bioengineering, 16*(1), 1–2. https://doi.org/10.1007/s12195-022-00754-8

Kung, T. H., Cheatham, M., Medenilla, A., Sillos, C., Leon, L. D., Elapaño, C., Madriaga, M., Aggabao, R., Diaz-Candido, G., Maningo, J., & Tseng, V. (2023). Performance of ChatGPT on USMLE: Potential for AI-assisted medical education using large language models. *PLOS Digital Health*, *2*(2), 1–12. https://doi.org/10.1371/journal.pdig.0000198

Liverpool, L. (2023). AI intensifies fight against 'paper mills' that churn out fake research. *Nature, 618*(7964), 222–223. https://doi.org/10.1038/d41586-023-01780-w

Lo, C. K. (2023). What is the impact of ChatGPT on education? A rapid review of the literature. *Education Sciences, 13*(4), 410–425. https://doi.org/10.3390/educsci13040410

Lund, B. D., & Naheem, K. T. (2024). Can ChatGPT be an author? A study of artificial intelligence authorship policies in top academic journals. *Learned Publishing, 37*(1), 13–21. https://doi.org/10.1002/leap.1582

Luo, J. (2024). A critical review of GenAI policies in higher education assessment: A call to reconsider the "originality" of students' work. *Assessment & Evaluation in Higher Education, 49*(1), 651–664. https://doi.org/10.1080/02602938.2024.2309963

McCabe, D. L. (1992). The influence of situational ethics on cheating among college students. *Sociological Inquiry, 62*(3), 365–374. https://doi.org/10.1111/j.1475-682X.1992.tb00287.x

Michel-Villarreal, R., Vilalta-Perdomo, E., Salinas-Navarro, D. E., Thierry-Aguilera, R., & Gerardou, F. S. (2023). Challenges and opportunities of generative AI for higher education as explained by ChatGPT. *Education Sciences, 13*(9), 856–874. https://doi.org/10.3390/educsci13090856

Moore, E. (2014). Accuracy of referencing and patterns of plagiarism in electronically published theses. *International Journal for Educational Integrity, 10*(1), 1–14. https://doi.org/10.21913/IJEI.v10i1.933

Moorhouse, B. L., Yeo, M. A., & Wan, Y. (2023). Generative AI tools and assessment: Guidelines of the world's top-ranking universities. *Computers and Education Open, 5*(15), 1–10. https://doi.org/10.1016/j.caeo.2023.100151

Nguyen, H. M., & Goto, D. (2024). Unmasking academic cheating behavior in the artificial intelligence era: Evidence from Vietnamese undergraduates. *Education and Information Technologies, 29*(1), 15999–16025. https://doi.org/10.1007/s10639-024-12495-4

Nguyen, K. V. (2025). The use of generative AI tools in higher education: Ethical and pedagogical principles. *Journal of Academic Ethics, 0*(0), 1–21. https://doi.org/10.1007/s10805-025-09607-1

Packalen, K. A., & Rowbotham, K. (2024). Clues to fostering a program culture of academic integrity: Findings from a multidimensional regression model. *International Journal for Educational Integrity, 20*(14), 1–27. https://doi.org/10.1007/s40979-024-00163-6

Perkins, M. (2023). Academic integrity considerations of AI Large Language Models in the post-pandemic era: ChatGPT and beyond. *Journal of University Teaching and Learning Practice, 20*(2), 1–26. https://doi.org/10.53761/1.20.02.07

Perkins, M., & Roe, J. (2024). Academic publisher guidelines on AI usage: A ChatGPT supported thematic analysis. *F1000Research, 12*(1398), 1–34. https://doi.org/10.12688/f1000research.142411.2

Perkins, M., Roe, J., Postma, D., McGaughran, J., & Hickerson, D. (2024). Detection of GPT-4 generated text in higher education: Combining academic judgement and software to identify generative AI tool misuse. *Journal of Academic Ethics, 22*(1), 89–113. https://doi.org/10.1007/s10805-023-09492-6

Piascik, P., & Brazeau, G. A. (2010). Promoting a culture of academic integrity. *American Journal of Pharmaceutical Education, 74*(6), 1–2. https://doi.org/10.5688/aj7406113

Prihar, A., & Wanna, C. (2019). Cheating at Yale: Students get honest about academic dishonesty. *Yale News*. https://yaledailynews.com/blog/2019/02/07/cheating-at-yale-students-get-honest-about-academic-dishonesty/. Accessed 8 May 2024.

Rahman, H., Islam, M. S., Andrabi, S. M. H., Sharma, J. P., & Reza, N. (2024). Comparing the similarity index across iThenticate, Ouriginal, and Turnitin plagiarism detection software. *Journal of Physical Education and Sport, 24*(2), 419–424. https://doi.org/10.7752/jpes.2024.02051

Robinson, E. M., & Tagher, C. G. (2017). The companion dissertation: Enriching the doctoral experience. *Journal of Nursing Education, 56*(9), 564–566. https://doi.org/10.3928/01484834-20170817-10

Schei, O. M., Møgelvang, A., & Ludvigsen, K. (2024). Perceptions and use of AI chatbots among students in higher education: A scoping review of empirical studies. *Education Sciences, 14*(1), 922–945. https://doi.org/10.3390/educsci14080922

Segbenya, M., Bervell, B., Frimpong-Manso, E., Otoo, I. C., Andzie, T. A., & Achina, S. (2023). Artificial intelligence in higher education: Modelling the antecedents of artificial intelligence usage and effects on 21st century employability skills among postgraduate students in Ghana. *Computers and Education: Artificial Intelligence, 5*(1), 1–13. https://doi.org/10.1016/j.caeai.2023.100188

Singh, S., & Remenyi, D. (2018). Plagiarism and ghostwriting: The rise in academic misconduct. *South African Journal of Science, 112*(5/6), 7. https://doi.org/10.17159/sajs.2016/20150300

Sweeney, S. (2023). Who wrote this? Essay mills and assessment – Considerations regarding contract cheating and AI in higher education. *The International Journal of Management Education, 21*(2), 100818. https://doi.org/10.1016/j.ijme.2023.100818

Tang, A., Li, K.-K., Kwok, K. O., Cao, L., Luong, S., & Tam, W. (2024). The importance of transparency: Declaring the use of generative artificial intelligence (AI) in academic writing. *Journal of Nursing Scholarship, 56*(2), 314–318. https://doi.org/10.1111/jnu.12938

Turnitin. (2024). *AI writing detection in the new, enhanced similarity report*. https://guides.turnitin.com/hc/en-us/articles/22774058814093-AI-writing-detection-in-the-new-enhanced-Similarity-Report. Accessed 10 Jan 2025.

Walters, W. H. (2023). The effectiveness of software designed to detect AI-generated writing: A comparison of 16 AI text detectors. *Open Information Science, 7*(1), 1–24. https://doi.org/10.1515/opis-2022-0158

Weber-Wulff, D., Anohina-Naumeca, A., Bjelobaba, S., Foltýnek, T., Guerrero-Dib, J., Popoola, O., Šigut, P., & Waddington, L. (2023). Testing of detection tools for AI-generated text. *International Journal for Educational Integrity, 19*(1), 1–39. https://doi.org/10.1007/s40979-023-00146-z

Young, S. N., Vanwye, W. R., Schafer, M. A., Robertson, T. A., & Poore, A. V. (2019). Factors affecting PhD student success. *International Journal of Exercise Science, 12*(1), 34–45. https://doi.org/10.70252/CEJT2520

Zhang, G., Zhang, T., Mao, S., Xu, Q., & Ma, X. (2023). Supervisors' ethical leadership and graduate students' attitudes toward academic misconduct. *PLoS One, 18*(4), 1–17. https://doi.org/10.1371/journal.pone.0283032

Zimba, O., & Gasparyan, A. Y. (2021). Plagiarism detection and prevention: A primer for researchers. *Reumatologia, 59*(3), 132–137. https://doi.org/10.5114/reum.2021.105974

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